

Paper 02 — Quotes and Design Notes

Working notes. Copy-paste fodder for main.typ.

“Berners-Lee made this explicit in his 2001 Scientific American article with James Hendler and Ora Lassila, laying out the vision for the Semantic Web. The article describes a web in which ‘information is given well-defined meaning, better enabling computers and people to work in cooperation.’ The core architecture they proposed — RDF for structured assertions, ontologies for defining relationships between concepts, URIs for ensuring that every concept ties back to a unique, discoverable definition — is a provenance architecture. It was designed so that ‘using a different URI for each specific concept’ would prevent semantic collapse, where a word means one thing in one context and another thing elsewhere. Berners-Lee, Hendler, and Lassila saw clearly what would happen without this layer. They wrote: ‘Human language thrives when using the same term to mean somewhat different things, but automation does not.’ Their solution was ontologies — formal definitions of concepts, relationships, and inference rules — that would allow machines to follow meaning back to its source, verify it, and reason about it. They envisioned agents that would exchange ‘proofs’ of their reasoning — machines that could show their work, trace their conclusions back to their premises, and let other machines verify the chain. The Semantic Web was provenance infrastructure for the digital age.” [1]

Note: RankeDB disagrees with the “one URI per concept” premise. Terms are ambiguous by nature; resolution is contextual and provenance-tracked, not globally prescribed. The ontology emerges from workers’ contextual interpretations rather than from a predefined global vocabulary. See §X on normalization and entity resolution.

Bibliography

[1] J. Talisman, “Where Provenance Ends, Knowledge Decays.”